SENIOR SOFTWARE ENGINEER

AI Systems · Backend Architecture · Production Infrastructure

Document: Technical Engineering Portfolio
Focus: AI Systems & Backend Architecture
Audience: CTOs, Engineering Managers & Founders
Specialization: Vibe Code → Production Ready

Executive Summary: Senior Software Engineer specializing in taking rapidly built or vibe-coded applications and re-engineering them into clean, reliable, production-ready systems. Deep expertise across FastAPI backend microservices, high-throughput vLLM model serving, adaptive RAG pipelines, enterprise Microsoft Entra ID / OAuth 2.0 security, and automated CI/CD.

1. ENGINEERING CASE STUDIES (FLAGSHIP WORK)

CASE STUDY 01: Enterprise LLM Application [Confidential Client Work]

Domain: Australian Engineering Consultancy (Cerecon) | Key Outcome: Turnaround reduced from Days → < 1 Hour

The Problem: Automated proposal generation needed to convert raw client briefs into audit-ready fee proposals with automated rate cards and branded Word/Excel deliverables without risking hallucinated pricing or audit gaps.

Engineering Implementation:
- **Backend Architecture:** Built and maintained FastAPI services on Azure App Service with strict domain separation.
- **Multi-Stage Prompt Assembly:** Engineered structured context preservation across multi-phase LLM runs.
- **Security & Identity:** Implemented per-user OAuth 2.0 token handshakes with Total Synergy v4 API and Microsoft Entra ID RBAC across 24 mapped enterprise use cases, eliminating shared-account audit gaps.
- **DevOps:** Solved Azure App Service limits using Kudu async zipdeploy with zero commit loss migration.

`Tech Stack: Python, FastAPI, Azure App Service, Azure DevOps, Microsoft Entra ID, OAuth 2.0, RBAC, REST APIs, LLMs`

CASE STUDY 02: High-Throughput LLM Inference Service [AI Infrastructure]

Hardware Target: Single NVIDIA RTX 2080 Ti | Key Outcome: 100 Concurrent Requests Supported

The Problem: Deploying production LLM serving on constrained GPU hardware while handling high concurrent user loads without VRAM out-of-memory crashes or high time-to-first-token (TTFT) queue spikes.

Engineering Implementation:
- **vLLM Engine Integration:** Leveraged continuous batching and PagedAttention to eliminate memory fragmentation.
- **KV-Cache Optimization:** Tuned memory allocation chunks and asynchronous request queue buffers.
- **Pipeline Flow:** Client → API Gateway → Async Queue / Scheduler → vLLM Batcher → GPU VRAM → Streamed Tokens.
- **Benchmarking:** Profiled token generation rates under synthetic load to maintain sub-second first-token response.

`Tech Stack: Python, vLLM, Continuous Batching, PagedAttention, PyTorch, FastAPI, Docker, GPU Profiling`

CASE STUDY 03: Contextual Dynamic RAG Framework [Search & LLM Infrastructure]

Architecture: Hybrid Dynamic Routing | Key Outcome: +60% Retrieval Efficiency Improvement

The Problem: Standard naive RAG pipelines suffer from high latency on exact-match queries and poor context quality on complex multi-hop reasoning over heterogeneous enterprise documents.

Engineering Implementation:
- **Dynamic Query Router:** Classifies incoming queries by intent, keyword density, and semantic complexity.
- **Hybrid Retrieval Layer:** Routes between FAISS dense vector search and BM25 sparse keyword indices dynamically.
- **Cross-Encoder Reranking:** Applied fine-grained reranking over candidate chunks before prompt context assembly.
- **Compaction:** Deduplicated context windows to maximize answer precision and minimize LLM inference tokens.

`Tech Stack: FAISS, BM25, Cross-Encoder, LangGraph, Python, FastAPI, Hugging Face Embeddings`

2. TECHNICAL STACK & CORE COMPETENCIES

Languages	Python, C++, SQL, JavaScript, HTML/CSS, Bash
AI / LLM Systems	LoRA / PEFT, Unsloth, vLLM, llama.cpp, GGUF, Dynamic RAG, Prompt Engineering, LangChain, LangGraph, Hugging Face Transformers, OpenAI / Anthropic / Gemini APIs, Coqui TTS
ML / Computer Vision	PyTorch, TensorFlow, TensorFlow Lite, Ultralytics YOLO (v8), OpenCV, scikit-learn, Optuna, Weights & Biases, Pandas, NumPy
Backend & Cloud	FastAPI, REST APIs, Docker, Git, Azure App Service, Azure DevOps, Microsoft Entra ID, OAuth 2.0, RBAC, CI/CD Pipelines, Hugging Face Spaces
Databases & Analytics	PostgreSQL, MySQL, MongoDB, Supabase, FAISS Vector Index, Power BI, Tableau

3. PUBLIC PROJECTS & OPEN-SOURCE ARTIFACTS

Kisan Rabta **Voice-to-Insight Agricultural AI Platform**	Fine-tuned OpenAI Whisper on 18,000 scraped text samples & 2 hours of regional Urdu audio. Extracted structured agricultural information & crop disease diagnostics via custom NER pipelines, exposed via low-latency REST APIs.	`Stack: Whisper, NER, Python, FastAPI, React Native`
Medical Reasoning LLM **DeepSeek R1 Distill LLaMA 8B**	Fine-tuned DeepSeek R1 Distill LLaMA 8B on clinical diagnostic reasoning datasets using Unsloth 4-bit QLoRA to maximize VRAM efficiency. Published model weights for public research and experimentation.	`Stack: DeepSeek, LLaMA, LoRA, Unsloth, PyTorch`
ID Document Parsing **Structured National Identity Extraction**	Computer vision and OCR pipeline for localized bounding box detection (Name, ID number, DOB, Barcode/QR) with automated cross-field checksum validation deployed to interactive web spaces.	`Stack: YOLOv8, OCR, OpenCV, Python, HF Spaces`
Automatic Number Plate Recognition (ANPR) **Real-Time Vehicle Localization**	High-accuracy object detection achieving **97.5% Precision and 97.5% Recall** on benchmark splits across challenging lighting and video stream angles.	`Stack: YOLOv8, OpenCV, Python, HF Spaces`

4. CLIENT ENGAGEMENTS & TECHNICAL TRACK RECORD

Enterprise AI & Cloud Systems **Australian Engineering Consultancy** **Jun 2026 — Present**	Lead engineering delivery on AI enterprise software, backend architecture, third-party integrations, and Azure deployments. Rebuilt prompt-assembly layer to reduce proposal turnaround from days to <1 hr; implemented Microsoft Entra ID RBAC across 24 enterprise use cases and per-user OAuth 2.0 with Total Synergy v4 API.
Applied AI & High-Throughput Serving **AI Engineering Consultancy** **Jun 2025 — Jun 2026**	Shipped client AI systems including vLLM inference microservices (100 concurrent requests on RTX 2080 Ti), dynamic RAG frameworks (+60% retrieval efficiency), edge computer vision deployments (TF Lite GPU), and quantized offline Android GGUF RAG chatbots.
Analytics & Applied Machine Learning **FinTech & Healthcare Engagements** **2023 — 2024**	Engineered real-time financial price prediction models for banking decision support, digital receipt OCR extraction pipelines (+15% extraction precision), and executive data analytics pipelines.
Technical Standards & Core Rigor **Computer Science Foundation**	**Bachelor of Science in Computer Science (B.S. CS)** — Rigorous foundation in Distributed Backend Systems, Scalable Software Architecture, Machine Learning, and Production Infrastructure.